FINAL REPORT – Movie Recommendation System (NLP)

Title

Content-Based Movie Recommendation System using TF-IDF and Cosine Similarity

Objective

The objective of this project is to develop **a movie Recommendation system** using Natural Language Processing (NLP) techniques.

The system recommends movies based on:

- Movie title
- User description (query)
- Genre-based filtering

Dataset

- Source: IMDB Genres dataset (Hugging Face – **jquigl/imdb-genres**)
- Fields used:
  - movie_name
  - description
  - genre
  - rating
  - votes
  - year
  - runtime

Data Preprocessing

The dataset was cleaned and prepared using the following steps:

Text Cleaning

- Converted text to lowercase

- Removed punctuation using regex
- Removed stopwords using nltk
- Handled missing values

Data Cleaning

- Removed duplicate movies
- Converted:
  - Year → integer
  - runtime → numeric
  - rating, votes → numeric

Vectorization

We used **TF-IDF (Term Frequency – Inverse Document Frequency)**.

- Max features: 8000
- N-grams: (1,2)
- Minimum document frequency: 3

Similarity Measurement

Used **Cosine Similarity** to measure similarity between movies:

- Computes angle between vectors
- Higher value = more similar movies

<u>Recommendation Methods</u>

Title-Based Recommendation

- Input: Movie title
- Process:
  - Find movie index
  - Compute similarity with all movies
  - Sort results based on:
    - Similarity score
    - Rating
    - Votes (log-scaled)
- Output: Top 5 similar movies

Description-Based Recommendation

- Input: User query (e.g., "space mission alien planet")
- Process:
  - Clean query
  - Convert to TF-IDF vector
  - Compute similarity with all movies
- Output: Top 5 similar movies

Genre-Based Recommendation

- Input: Genre + query
- Process:
  - Filter dataset by genre
  - Apply TF-IDF on filtered data
  - Compute similarity
- Output: Top 5 movies in that genre

Sample Outputs

```
print("\n TITLE TEST:")
print(recommend_by_title("avatar: the way of water"))

print("\n DESCRIPTION TEST:")
print(recommend_by_description("space mission alien planet survival"))

print("\n GENRE TEST:")
print(recommend_by_genre("action", "hero saves world"))
```

```
 TITLE TEST:
['the arizona terror', 'poor agnes', 'long distance', 'nu hou kuang hua', 'moss rose']

 DESCRIPTION TEST:
['morpheus one', "project 'gemini'", 'space chimps', 'blizkata dalechina', 'fate federal agent 8']

 GENRE TEST:
['nordexpressen', 'desert raiders', 'terror of the steppes', 'guns of the apocalypse', 'beyond the game']
```

Improvements Implemented

Combined:

- Cosine similarity
- Movie ratings
- Vote counts (log scaled)

This improves recommendation quality significantly.

Limitations

- Only uses textual data (no user history)
- Cannot recommend new/unseen movies
- Genre filtering depends on dataset quality
- TF-IDF does not capture deep meaning (context)

Future Enhancements

- Use Deep Learning models (BERT, Transformers)
- Add collaborative filtering
- Use user preferences/history
- Deploy as web app (Streamlit / Flask)
- Improve speed using approximate nearest neighbors

Evaluation Metrics

- Cosine similarity score
- Manual validation of recommendations
- Relevance of recommended movies

Technologies Used

- Python
- Pandas
- Scikit-learn
- NLTK

Conclusion

This project successfully builds a **content-based movie recommendation system** using NLP techniques.
The system provides accurate recommendations using:

- TF-IDF vectorization
- Cosine similarity
- Rating and popularity weighting